

Data-driven adaptive optimal output tracking with assured convergence rate

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ARTICLE INFO

Keywords:

Optimal output tracking
Assured convergence rate
Adaptive dynamic programming
Continuous-time linear system

ABSTRACT

This paper addresses the optimal output tracking problem for a class of systems with unknown dynamics and specified convergence rates. We propose an adaptive optimal control method based on the adaptive dynamic programming (ADP) method, which solely relies on input-output data. Specifically, we re-express the system state based on input-output data and design the dynamic feedback controller accordingly. Subsequently, we propose two ADP algorithms based on policy iteration and value iteration to determine the optimal control gain and provide the conditions for ensuring the convergence of both algorithms under a given tracking error convergence rate. Finally, the simulation study results demonstrate the proposed method's effectiveness.

1. Introduction

The output tracking problem addresses the design of a feedback controller that achieves asymptotic tracking of a class of reference inputs while ensuring the stability of the closed-loop system. This represents a general mathematical formulation with broad practical implications across various domains, such as aircraft [1], satellites [2], robotic manipulators [3], and vehicles [4]. It is common practice to achieve optimal output tracking performance using the linear quadratic regulator (LQR) to determine the optimal gain for state feedback control. A critical step in this process involves solving the algebraic Riccati equation (ARE), an area of significant research interest [5–7]. Amongst various methods employed, the policy iteration (PI) algorithm proposed by Kleinman [8] remains one of the most widely used approaches. However, solving the output regulation equation and using LQR for optimal controller design presuppose precise knowledge of the system dynamics. This idealized assumption can be challenging to satisfy in practical applications.

Alternative methods that do not require complete knowledge of system dynamics have been explored to address the above limitation. In recent years, methods based on machine learning and model-free adaptive control have become one of the research hotspots [9–17]. However, methods based on adaptive dynamic programming (ADP) have remained popular, particularly following the work of Jiang et al. [18] and Bian et al. [19], who built upon the PI algorithm and value iteration (VI) algorithm by incorporating the concept of ADP. Their methods enable iterative solutions to the ARE using only online state and input data, thereby eliminating the assumption of knowledge of system dynamics. Gao et al. [20] applied this method to the problem of optimal output tracking to find the optimal solution for the output regulation equation and the ARE for CT systems. Jiang et al. [21] extended this method

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<https://doi.org/10.1016/j.jfranklin.2025.108086>

Received 21 October 2024; Received in revised form 21 July 2025; Accepted 16 September 2025

Available online 24 September 2025

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to discrete-time (DT) systems and developed a VI version to broaden its applicability [22]. Additionally, Xu et al. [23] devised a mixed iteration algorithm that combines the advantages of VI and PI algorithms to address distributed optimal tracking control for multi-agent systems without requiring an initial stabilizing control gain and exhibiting faster convergence compared to traditional VI algorithms. Furthermore, Xie et al. [24] designed an enhanced virtual system to replace the original system and its internal model, combining ADP methods with data from this enhanced virtual system to propose a new iterative learning equation to achieve optimal output regulation.

These approaches effectively circumvent the necessity for a system dynamics but typically presuppose accessibility to the system state. For instance, in [24], not only the system state but also its derivatives are required to construct the enhanced virtual system. This implies that practical applications necessitate real-time capability for measuring the system state variables.

Numerous researchers have devoted significant efforts to addressing the LQR problem in cases where the system dynamics are unknown, and the states are not directly accessible. Lewis et al. [25] employed delayed input-output data to re-express the state of linear DT systems. Rizvi et al. [26] utilized this state re-expression method to implement a model-free control scheme for optimal tracking. Similarly, Zhao et al. [27] re-expressed the state of CT systems using delayed output data. Despite these advancements, it has been noted that the method proposed by Lewis et al. [25] exhibits deadbeat characteristics [28], and the method in [27] may lead to estimation bias after injecting exploration noise [29].

A different category of approaches involves the re-expression of system states using state observers. Zhu et al. [30] employed an adaptive state observer to acquire knowledge of the full state, but this method requires partial knowledge of system dynamics and can only converge to a suboptimal solution. Recently, Rizvi et al. [31] introduced a state parameterization method derived from the Luenberger observer design formula and the corresponding ADP method, which was subsequently extended to CT systems in [29]. The extension of this approach to the output tracking problem has generated significant interest [32–37] as it offers a practical solution to dealing with unknown system dynamics and inaccessible states. However, it is essential to note that this method may result in re-expressed states with high dimensionality, requiring more data collection and computational effort during the iterative process. Furthermore, these methods require sufficient running time for the system before applying the ADP method to ensure that re-expression errors are sufficiently small [32], posing challenges when combined with the VI algorithm due to potential divergence caused by initial control gains. Additionally, beyond asymptotic tracking, the convergence rate of closed-loop systems is also crucial in practical applications.

In this paper, we address the problem of achieving optimal output tracking control with a specified convergence rate. We propose a data-driven dynamic feedback controller and two methods for designing such a controller, which utilize system input-output data to determine the optimal control gain without relying on the system dynamics and states and guarantee optimal output tracking control at a specified convergence rate. The primary contributions of this study encompass:

- (1) Propose a method of state re-expression based on reduced-order observer and prove the asymptotic equivalence relationship between the re-expressed state and the actual state. Compared to the method in [29], less computational effort is needed, and the dimension of the state after re-expression is smaller.
- (2) Design a dynamic output feedback controller and propose an algorithm based on PI algorithms within the ADP framework for computing optimal control gain with assured convergence rate. Compared with the method of [32], which requires two exploration noises, the proposed algorithm only needs one and requires less data to be collected.
- (3) Present the VI version of the above method, which overcomes the stringent assumption of the initial stabilizing control. Different from the method in [29], this algorithm can use output feedback control as the initial control policy, thus alleviating the problem of system output divergence.

The remaining sections of the paper are structured as follows: Section 2 introduces the problem statement and preliminary concepts. Section 3 explains the state re-expression method, analyses the re-expression error, and presents two corresponding ADP algorithms. Section 4 validates the effectiveness of these algorithms through simulations. Finally, conclusions drawn from this study are presented in Section 5.

Notations: Throughout this paper, $\mathbb{Z}$ denotes the set of integers, $\mathbb{C}$ represents the complex plane, $\mathbb{C}_-$ stands for the open left half of the complex plane, I_n and $0_{n \times m}$ denote respectively the $n \times n$ identity matrix and the $n \times m$ zero matrix, $\mathcal{P}^n$ indicates the set of $n \times n$ real symmetric matrices, $\mathcal{P}_+^n = \{P \in \mathcal{P}^n : P \geq 0\}$, and $\otimes$ denotes the Kronecker product. For a matrix $X \in \mathbb{R}^{n \times n}$, $X > (\geq) 0$ means that the matrix X is positive (semi-positive), $\sigma(X)$ represents the spectrum of matrix X , and $\text{vecs}(X) = [x_{11}, 2x_{12}, \dots, 2x_{1n}, x_{22}, 2x_{23}, \dots, 2x_{n-1,n}, x_{nn}]^T \in \mathbb{R}^{\frac{1}{2}n(n+1)}$. For a matrix $X \in \mathbb{R}^{n \times m}$, $\text{vec}(X) = [x_1^T, x_2^T, \dots, x_m^T]^T \in \mathbb{R}^{nm}$, where $x_i \in \mathbb{R}^n$ is the i th column of X . For a vector $v \in \mathbb{R}^n$, $\text{vecv}(v) = [v_1^2, v_1 v_2, \dots, v_1 v_n, v_2^2, v_2 v_3, \dots, v_{n-1} v_n, v_n^2]^T \in \mathbb{R}^{\frac{1}{2}n(n+1)}$.

2. Problem formulation and preliminaries

2.1. Problem formulation

Consider a class of CT linear time-invariant systems described as

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the system state, $u(t) \in \mathbb{R}^m$ and $y(t) \in \mathbb{R}^p$ are the control input and the output, respectively. The matrices $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, and $C \in \mathbb{R}^{p \times n}$ are unknown constant matrices.

Following the output regulation framework, the reference signal is assumed to be generated by a linear autonomous differential equation as follows:

$$\begin{cases} \dot{v}(t) = Dv(t) \\ r(t) = -Ev(t) \end{cases} \quad (2)$$

where $v(t) \in \mathbb{R}^q$ and $r(t) \in \mathbb{R}^p$ are the exosystem state and the reference signal, respectively. The matrices $D \in \mathbb{R}^{q \times q}$ and $E \in \mathbb{R}^{p \times q}$ are constant matrices.

Thus, the tracking error is given by

$$e(t) = Cx(t) + Ev(t) \quad (3)$$

For systems (1) and (2), we list the following assumptions:

Assumption 1 ([38]). The pair (A, B) is stabilizable.

Assumption 2 ([38]). The matrix D has no eigenvalues with negative real parts.

Assumption 3 ([38]). $\text{rank} \left(\begin{bmatrix} A - \lambda I_n & B \\ C & 0_{p \times m} \end{bmatrix} \right) = n + p, \forall \lambda \in \sigma(D)$.

Assumption 4. The pair (A, C) is observable.

Remark 1. Assumptions 1–3 are standard assumptions for the linear output regulation problem and can be found in [38]. Assumption 4 is necessary to enable the output data to be used for re-expressing the system state in cases where the system state is not available.

We consider the following dynamic feedback controller:

$$\begin{cases} \dot{z}(t) = G_1 z(t) + G_2 e(t) \\ u(t) = -K_x x(t) - K_z z(t) \end{cases} \quad (4)$$

where the matrix pair (G_1, G_2) with $G_1 \in \mathbb{R}^{pq \times pq}$, and $G_2 \in \mathbb{R}^{pq \times p}$ is assumed to incorporate a minimal p -copy internal model of matrix D . $z(t) \in \mathbb{R}^{pq}$ is the state of the incorporate internal model. $K_x \in \mathbb{R}^{m \times n}$ and $K_z \in \mathbb{R}^{m \times pq}$ represent the control gain matrices.

The resulting closed-loop system formed by combining systems (1) and (2) is described as follows:

$$\begin{cases} \dot{x}_c(t) = \bar{A}x_c(t) + \bar{B}u(t) + \bar{G}v(t) \\ e(t) = \bar{C}x_c(t) + Ev(t) \end{cases} \quad (5)$$

where $x_c(t) = [x^\top(t), z^\top(t)]^\top$, $\bar{A} = \begin{bmatrix} A & 0_{n \times pq} \\ G_2 C & G_1 \end{bmatrix}$, $\bar{B} = \begin{bmatrix} B \\ 0_{pq \times m} \end{bmatrix}$, $\bar{G} = \begin{bmatrix} 0_{n \times q} \\ G_2 E \end{bmatrix}$, and $\bar{C} = [C, 0_{p \times pq}]$. Then, the controller can be written as $u(t) = -\bar{K}x_c(t)$ with $\bar{K} = [K_x, K_z]$.

To proceed further, we introduce the following result:

Lemma 1 ([38]). Under Assumptions 1–3, if the pair (G_1, G_2) incorporates a p -copy internal model of matrix D , then the pair

$$\left(\begin{bmatrix} A & 0_{n \times pq} \\ G_2 C & G_1 \end{bmatrix}, \begin{bmatrix} B \\ 0_{pq \times m} \end{bmatrix} \right)$$

is stabilizable.

According to Lemma 1, the matrix pair $(\bar{A}, \bar{B})$ in the system (5) is controllable. It implies that if the matrix $\bar{K}$ is chosen appropriately, then $(\bar{A} - \bar{B}\bar{K})$ can be Hurwitz. Moreover, there exists a unique matrix X satisfying the following equation [38]:

$$\begin{cases} \bar{G} = XD - (\bar{A} - \bar{B}\bar{K})X \\ E = -\bar{C}X \end{cases} \quad (6)$$

By defining the new variable $\bar{x}_c(t) = x_c(t) - Xv(t)$, the system (5) can be equivalently expressed as follows:

$$\begin{cases} \dot{\bar{x}}_c(t) = \bar{A}\bar{x}_c(t) + \bar{B}\bar{u}(t) \\ e(t) = \bar{C}\bar{x}_c(t) \end{cases} \quad (7)$$

where $\bar{u}(t) = -\bar{K}\bar{x}_c(t)$.

Our aim is to design an optimal controller that ensures the tracking error $e(t)$ satisfies the following equation:

$$\lim_{t \rightarrow \infty} e^{\gamma t} e(t) = 0 \quad (8)$$

where the convergence rate criterion γ satisfies $0 \leq \gamma < \infty$.

Under the control goal delineated in Eq. (8), we introduce transformed variables for the system state and tracking error. Specifically, let $\bar{x}'_c(t) = e^{t'} \bar{x}_c(t)$ and $e'(t) = e^{t'} e(t)$. By doing so, we transform the system (7) into a new one, described as follows:

$$\begin{cases} \dot{\bar{x}}'_c(t) = \bar{A}_\gamma \bar{x}'_c(t) + \bar{B} \bar{u}'(t) \\ e'(t) = \bar{C} \bar{x}'_c(t) \end{cases} \quad (9)$$

where $\bar{A}_\gamma = \bar{A} + \gamma I_{n+pq}$ and $\bar{u}'(t) = -\bar{K} \bar{x}'_c(t)$.

From Lemma 1, it is known that the pair $(\bar{A}, \bar{B})$ is stabilizable, and according to the PBH test, it follows that

$$\text{rank}([\bar{A} - \lambda I_{n+pq}, \bar{B}]) = n + pq \quad \forall \lambda \in \mathbb{C}$$

Then, the following can be derived:

$$\text{rank}([\bar{A}_\gamma - (\lambda + \gamma)I_{n+pq}, \bar{B}]) = n + pq \quad \forall \lambda \in \mathbb{C}$$

Therefore, the matrix pair $(\bar{A}_\gamma, \bar{B})$ in the system (9) is stabilizable.

Building upon this foundation, to achieve optimal tracking with a specified convergence rate, the control gain $\bar{K}$ is designed based on the following optimal control problem:

Problem 1.

$$\begin{aligned} \min_{\bar{u}'(\cdot)} \quad & \int_{t_0}^{\infty} (\bar{x}'_c{}^T(\tau) Q \bar{x}'_c(\tau) + \bar{u}'^T(\tau) R \bar{u}'(\tau)) d\tau \\ \text{s.t.} \quad & \dot{\bar{x}}'_c(t) = \bar{A}_\gamma \bar{x}'_c(t) + \bar{B} \bar{u}'(t) \end{aligned} \quad (10)$$

where $Q = Q^T \geq 0$, $R = R^T > 0$, and the pair $(\bar{A}_\gamma, \sqrt{Q})$ is observable. In this paper, the matrix Q is constructed as $Q = \begin{bmatrix} C^T Q_e C & 0_{n \times pq} \\ 0_{pq \times n} & Q_z \end{bmatrix}$ with $Q_e > 0$ and $Q_z > 0$. The matrix Q_z is designed to ensure that the pair $(\bar{A}_\gamma, \sqrt{Q})$ is observable.

By defining $\bar{Q}_z = \begin{bmatrix} 0_{n \times n} & 0_{n \times pq} \\ 0_{pq \times n} & Q_z \end{bmatrix}$, we reformulate the quadratic form of the state $\bar{x}'_c(t)$ as follows:

$$\bar{x}'_c{}^T(t) Q \bar{x}'_c(t) = e'^T(t) Q_e e'(t) + \bar{x}'_c{}^T(t) \bar{Q}_z \bar{x}'_c(t) \quad (11)$$

Remark 2. The influence of matrices Q_e , Q_z , and R on the convergence rate of the system (7) is implicit and difficult to quantify directly. Introducing the parameter γ explicitly specifies the exponential convergence rate of the tracking error, thereby addressing this limitation.

2.2. Preliminaries

Problem 1 is an instance of the LQR problem. According to linear optimal control theory, the optimal control gain $\bar{K}^*$ satisfies the following equation:

$$\bar{K}^* = R^{-1} \bar{B}^T P^* \quad (12)$$

Here, the matrix P^* , which satisfies $P^* = P^{*\top} > 0$, is the unique positive definite solution to the following ARE:

$$\bar{A}_\gamma^T P^* + P^* \bar{A}_\gamma + Q - P^* \bar{B} R^{-1} \bar{B}^T P^* = 0 \quad (13)$$

If the system dynamics are known, Eq. (13) can be solved using PI and VI algorithms, which are recalled below.

Lemma 2 ([8]). Let $\bar{K}^0 \in \mathbb{R}^{m \times (n+pq)}$ be any stabilizing control gain, and the matrix P^k be the solution of Lyapunov equation:

$$(\bar{A}_\gamma - \bar{B} \bar{K}^k)^T P^k + P^k (\bar{A}_\gamma - \bar{B} \bar{K}^k) + Q + (\bar{K}^k)^T R \bar{K}^k = 0 \quad (14)$$

where for each $k = 1, 2, \dots$,

$$\bar{K}^k = R^{-1} \bar{B}^T P^{k-1} \quad (15)$$

Then, the following properties hold:

- 1) $\sigma(\bar{A}_\gamma - \bar{B} \bar{K}^k) \subset \mathbb{C}_-;$
- 2) $P^* \leq P^{k+1} \leq P^k;$
- 3) $\lim_{k \rightarrow \infty} P^k = P^*, \lim_{k \rightarrow \infty} \bar{K}^k = \bar{K}^*.$

Lemma 3 ([19]). Let $P^0 \in \mathbb{R}^{(n+pq) \times (n+pq)}$ be any real symmetric positive definite matrix and $i = 0$. P^{k+1} is the solution of following equation:

$$\begin{cases} \tilde{P}^{k+1} = P^k + \epsilon^k (\bar{A}_\gamma^T P^k + P^k \bar{A}_\gamma + Q - P^k \bar{B} R^{-1} \bar{B}^T P^k) \\ P^{k+1} = \begin{cases} P^0 & \tilde{P}^{k+1} \notin B^i \\ \tilde{P}^{k+1} & \tilde{P}^{k+1} \in B^i \end{cases} \end{cases} \quad (16)$$

where the step size sequence ϵ^k satisfies $\lim_{k \rightarrow \infty} \epsilon^k = 0$. $\{B^i\}_{i=0}^\infty$ is a collection of bounded sets with nonempty interiors which satisfy $B^i \subseteq B^{i+1}$ and $\lim_{i \rightarrow \infty} B^i = \mathcal{P}_+^n$. The variable i is determined by the following equation:

$$i = \begin{cases} i+1 & \tilde{P}^k \notin B^i \\ i & \tilde{P}^k \in B^i \end{cases}$$

Then, one has $\lim_{k \rightarrow \infty} P^k = P^*$.

In this paper, since the system state $x(t)$ are not directly accessible, we propose a state re-expression method based on the following reduced-order observer in the subsequent sections.

$$\begin{cases} \dot{w}(t) = Fw(t) + Jy(t) + Hu(t) \\ \hat{x}(t) = Q_1y(t) + Q_2w(t) \end{cases} \quad (17)$$

where $w(t) \in \mathbb{R}^{n_w}$ with $n_w = n - p$ represents the observer state. $\hat{x}(t) \in \mathbb{R}^n$ is the reconstructed state. $F \in \mathbb{R}^{n_w \times n_w}$, $J \in \mathbb{R}^{n_w \times p}$, $H \in \mathbb{R}^{n_w \times n}$, $Q_1 \in \mathbb{R}^{n \times p}$ and $Q_2 \in \mathbb{R}^{n \times n_w}$ are observer matrices.

Lemma 4 ([39]). Under *Assumption 4*, the necessary and sufficient conditions for Eq. (17) to serve as a state observer for the system (1) are as follows:

- 1) There exists a full-rank matrix $T \in \mathbb{R}^{n_w \times n}$ such that $L = \begin{bmatrix} C \\ T \end{bmatrix}$ is nonsingular;
- 2) All eigenvalues of matrix F have negative real parts;
- 3) $TA - FT = JC$;
- 4) $H = TB$;
- 5) $[Q_1, Q_2] = L^{-1}$.

Lemma 5 ([39]). For the system (1) and the reduced-order observer 17, assuming matrices A and F do not have any common eigenvalues, a necessary condition for the sylvester equation $TA - FT = JC$ to admit a full-rank solution matrix T rendering $L = \begin{bmatrix} C \\ T \end{bmatrix}$ nonsingular is that the pair (A, C) is observable and the pair (F, J) is controllable.

Remark 3. To determine the matrices involved in the state observer defined by Eq. (17), knowledge of the dynamics of the system (1) remains essential. We address this issue in subsequent sections by further developing methods for re-expressing the system state based on Eq. (17).

3. Main results

3.1. State re-expression based on input-output data

Applying the Laplace transformation to Eq. (17) yields

$$\begin{aligned} w(s) &= (sI_{n_w} - F)^{-1} (Jy(s) + Hu(s) + w(0)) \\ &= \frac{M(s)}{a(s)} (Jy(s) + Hu(s) + w(0)) \end{aligned} \quad (18)$$

where $a(s) = s^{n_w} + a_1 s^{n_w-1} + \dots + a_{n_w-1} s + a_{n_w}$ represents the characteristic polynomial of matrix F . $M(s) = M_1 s^{n_w-1} + \dots + M_{n_w-1} s + M_{n_w}$ is the adjugate matrix of the characteristic matrix $(sI_{n_w} - F)$. $a(s)$ and $M(s)$ satisfy the following relationship:

$$\begin{aligned} a(s)I_{n_w} &= M(s)(sI_{n_w} - F) \\ &= M_1 s^{n_w} + (M_2 - M_1 F)s^{n_w-1} + \dots + (M_{n_w} - M_{n_w-1} F)s - M_{n_w} F \end{aligned}$$

Then

$$M_i = \begin{cases} I_{n_w} & i = 1 \\ M_{i-1}F + a_{i-1}I_{n_w} & i = 2, \dots, n_w \end{cases}$$

The $\frac{M(s)}{a(s)}Hu(s)$ in Eq. (18) can be rewritten as

$$\begin{aligned} \frac{M(s)}{a(s)}Hu(s) &= \frac{1}{a(s)} (M_1 H s^{n_w-1} + \dots + M_{n_w-1} H s + M_{n_w} H) u(s) \\ &= [M_{n_w} H, \dots, M_1 H] \begin{bmatrix} s^0 \\ \vdots \\ s^{n_w-1} \end{bmatrix}^\top \otimes \frac{u(s)}{a(s)} \\ &= \bar{M}_H \zeta_u(s) \end{aligned} \quad (19)$$

where $\bar{M}_H = [M_{n_w} H, M_{n_w-1} H, \dots, M_1 H] \in \mathbb{R}^{n_w \times n_w m}$ and $\zeta_u(s) = [s^0, \dots, s^{n_w-1}]^\top \otimes \frac{u(s)}{a(s)}$.

Let $A_\zeta = \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ -a_{n_w} & -a_{n_w-1} & \dots & -a_1 \end{bmatrix} \in \mathbb{R}^{n_w \times n_w}$, $b = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^{n_w}$. Then

$$\begin{aligned} s\zeta_u(s) &= [s, \dots, s^{n_w}]^\top \otimes \frac{u(s)}{a(s)} \\ &= \left(A_\zeta [s^0, \dots, s^{n_w-1}]^\top + [0, \dots, 0, a(s)]^\top \right) \otimes \frac{u(s)}{a(s)} \\ &= (I_m \otimes A_\zeta) \zeta_u(s) + (I_m \otimes b) u(s) \end{aligned}$$

If $\zeta_u(0) = 0_{n_w m \times 1}$, applying the inverse Laplace transform to the above equation yields

$$\dot{\zeta}_u(t) = (I_m \otimes A_\zeta) \zeta_u(t) + (I_m \otimes b) u(t) \quad (20)$$

For $\frac{M(s)}{a(s)} J y(s)$, we do not handle it in the form of Eq. (20) as we do for $\frac{M(s)}{a(s)} H u(s)$, but rather consider the following equation:

$$\dot{w}_1(t) = F w_1(t) + J y(t) \quad (21)$$

Similarly, applying the Laplace transform to Eq. (21) yields

$$w_1(s) = \frac{M(s)}{a(s)} J y(s) + \frac{M(s)}{a(s)} w_1(0)$$

Then, the following can be derived:

$$\frac{M(s)}{a(s)} J y(s) = w_1(s) - \frac{M(s)}{a(s)} w_1(0) \quad (22)$$

Based on Eqs. (18)–(22), it can be derived that

$$\begin{aligned} \hat{x}(t) &= Q_1 y(t) + Q_2 w(t) \\ &= [Q_1, Q_2, Q_2 \bar{M}_H] [y^\top(t), w_1^\top(t), \zeta_u^\top(t)]^\top \\ &\quad + Q_2 e^{Ft} (w(0) - w_1(0)) \\ &= M_Q \zeta(t) + Q_2 e^{Ft} (w(0) - w_1(0)) \end{aligned} \quad (23)$$

where $M_Q = [Q_1, Q_2, Q_2 \bar{M}_H] \in \mathbb{R}^{n \times (n+n_w m)}$, $\zeta(t) = [y^\top(t), w_1^\top(t), \zeta_u^\top(t)]^\top$. $M_Q \zeta(t)$ is utilized to re-express the state $x(t)$.

For Eq. (23), we present the following result:

Lemma 6. Based on Lemma 4, for the state re-expression as shown in Eq. (23), the following statements holds.

1) The re-expression error is given by

$$\begin{aligned} \tilde{x}(t) &= x(t) - M_Q \zeta(t) \\ &= Q_2 e^{Ft} (T x(0) - w_1(0)) \end{aligned} \quad (24)$$

2) $M_Q \zeta(t)$ and $x(t)$ satisfy the following asymptotic equivalence relation:

$$\lim_{t \rightarrow \infty} M_Q \zeta(t) = \lim_{t \rightarrow \infty} x(t) \quad (25)$$

Proof. By employing the conditions stipulated in Lemma 4, we have

$$\begin{aligned} \dot{w}(t) - T \dot{x}(t) &= (F w(t) + J y(t) + H u(t)) - T (A x(t) + B u(t)) \\ &= (J C - T A) x(t) + (H - T B) u(t) + F w(t) \\ &= F(w(t) - T x(t)) \end{aligned}$$

Solving the aforementioned equation yields

$$w(t) - T x(t) = e^{Ft} (w(0) - T x(0))$$

On this basis, the re-expressed state error can be derived.

$$\begin{aligned} \tilde{x}(t) &= (x(t) - \hat{x}(t)) + (\hat{x}(t) - M_Q \zeta(t)) \\ &= \left(x(t) - L^{-1} [y^\top(t), w_1^\top(t)]^\top \right) + Q_2 e^{Ft} (w(0) - w_1(0)) \\ &= L^{-1} \left(L x(t) - [y^\top(t), w_1^\top(t)]^\top \right) + Q_2 e^{Ft} (w(0) - w_1(0)) \\ &= Q_2 (T x(t) - w(t)) + Q_2 e^{Ft} (w(0) - w_1(0)) \\ &= Q_2 e^{Ft} (T x(0) - w(0)) + Q_2 e^{Ft} (w(0) - w_1(0)) \\ &= Q_2 e^{Ft} (T x(0) - w_1(0)) \end{aligned}$$

From Lemma 4, it is known that all eigenvalues of matrix F have negative real parts. Therefore, as $t \rightarrow \infty$ and $e^{Ft} \rightarrow 0_{n_w \times n_w}$. It can be proven that Eq. (25) holds. This completes the proof. $\square$

Employing Lemma 6, we proceed to re-express the state $x_c(t)$ of the system (5).

$$\begin{aligned} x_c(t) &= \begin{bmatrix} M_Q & 0_{n \times pq} \\ 0_{pq \times (n+n_wm)} & I_{pq} \end{bmatrix} \begin{bmatrix} \zeta(t) \\ z(t) \end{bmatrix} + \begin{bmatrix} \tilde{x}(t) \\ 0_{pq \times 1} \end{bmatrix} \\ &= \bar{M}_{\bar{\zeta}} \bar{\zeta}(t) + \omega(t) \end{aligned} \quad (26)$$

where $\bar{M}_{\bar{\zeta}} \in \mathbb{R}^{(n+pq) \times n_{\bar{\zeta}}}$ with $n_{\bar{\zeta}} = n + pq + n_wm$, $\bar{\zeta}(t) \in \mathbb{R}^{n_{\bar{\zeta}}}$, and $\omega(t) \in \mathbb{R}^{n+pq}$.

Furthermore, we analyze the impact of the state re-expression on the state feedback control.

Theorem 1. Considering the system (5), if $\bar{K}$ is a stabilizing control gain, the state driven by the state feedback controller $u_{c1}(t) = -\bar{K}x_c(t)$ asymptotically converges to the state driven by the re-expressed state feedback controller $u_{c2}(t) = -\bar{K}\bar{M}_{\bar{\zeta}}\bar{\zeta}(t)$.

Proof. Denote the states driven by $u_{c1}(t)$ and $u_{c2}(t)$ as $x_{c1}(t)$ and $x_{c2}(t)$, respectively. Then, it follows that

$$\begin{aligned} \dot{e}_{x_c}(t) &= \dot{x}_{c1}(t) - \dot{x}_{c2}(t) \\ &= (\bar{A} - \bar{B}\bar{K})x_{c1}(t) - ((\bar{A} - \bar{B}\bar{K})x_{c2}(t) + \bar{B}\bar{K}\omega(t)) \\ &= (\bar{A} - \bar{B}\bar{K})(x_{c1}(t) - x_{c2}(t)) - \bar{B}\bar{K}\omega(t) \\ &= (\bar{A} - \bar{B}\bar{K})e_{x_c}(t) - \bar{B}\bar{K}\omega(t) \end{aligned}$$

Since $(\bar{A} - \bar{B}\bar{K})$ is Hurwitz, and $\omega(t)$ converges zero asymptotically, according to the perturbation theory, e_{x_c} is asymptotically stable. This completes the proof. $\square$

Based on the conclusion of Theorem 1, the controller for the system (5) is redesigned as follows:

$$\begin{cases} \dot{z}(t) = G_1 z(t) + G_2 e(t) \\ u(t) = -\bar{K}_M \bar{\zeta}(t) \end{cases} \quad (27)$$

where $\bar{K}_M = \bar{K}\bar{M}_{\bar{\zeta}} \in \mathbb{R}^{m \times n_{\bar{\zeta}}}$.

Remark 4. In this subsection, the state $x(t)$ is re-expressed as a product of an unknown matrix M_Q and a vector $\zeta(t)$. All matrices in Eq. (17) related to the dynamics of the system (1) are included in M_Q . The vector $\zeta(t)$ is computed from input-output data using Eqs. (20) and (21). Furthermore, under the conditions stipulated in Lemma 5 for the pair (F, J) , matrix F shares no eigenvalues with matrix A , the sylvester equation $TA - FT = JC$ yields a unique full-rank solution matrix T . Consequently, the uniqueness of matrices L and M_Q can be established.

Remark 5. Note that ensuring no overlapping eigenvalues between A and F fundamentally relies on appropriately selecting F during the algorithm design phase. In practice, the eigenvalues of F can be designated as distinct negative real numbers or complex numbers with negative real parts, while ensuring that their magnitudes or real parts substantially exceed any expected range of the eigenvalues of A .

3.2. PI-Based ADP algorithm using input-output data

The state $\bar{x}_c(t)$ in the system (9) requires X derived from system dynamics, making both X and $\bar{x}_c(t)$ inexpressible. While Eq. (23) proposes state re-expression, this method fails to re-express $\bar{x}_c(t)$ due to the unobservability of $(\bar{A}_y, \bar{C})$. Given this limitation, we observe from Eq. (13) that the optimal control gain $\bar{K}^*$ depends solely on the model of the system (9), motivating our design of the following controller to leverage this dependency:

$$\begin{cases} \dot{z}_1(t) = G_1 z_1(t) + G_2 y(t) \\ u_1(t) = -\bar{K}_M \bar{\zeta}_1(t) + \xi_1(t) \end{cases} \quad (28)$$

where $\bar{\zeta}_1(t) = [y^\top(t), w_1^\top(t), \zeta_{u_1}^\top(t), z_1^\top(t)]^\top$, and $\zeta_{u_1}(t)$ follows $\zeta_u(t)$ in Eq. (20). $\xi_1(t) \in \mathbb{R}^m$ represents the bounded exploration noise.

By applying the controller (28) to the system (1), the closed-loop system can be described as follows:

$$\begin{cases} \dot{\rho}(t) = \bar{A}\rho(t) + \bar{B}u_1(t) \\ y(t) = \bar{C}\rho(t) \end{cases} \quad (29)$$

where $\rho(t) = [x^\top(t), z_1^\top(t)]^\top$.

Following a procedure analogous to that used for Eq. (9), we can derive a system from Eq. (29) which has the same model with Eq. (9).

$$\begin{cases} \dot{\rho}'(t) = \bar{A}_y \rho'(t) + \bar{B}u_1'(t) \\ y'(t) = \bar{C} \rho'(t) \end{cases} \quad (30)$$

where $\rho'(t) = e^{\gamma t} \rho(t)$, $y'(t) = e^{\gamma t} y(t)$, and $u'_1(t) = e^{\gamma t} u_1(t)$.

From Eq. (26), the state $\rho'(t)$ can be re-expressed as

$$\rho'(t) = \bar{M}_{\bar{\xi}} \bar{\xi}'_1(t) + \omega'(t) \quad (31)$$

where $\bar{\xi}'_1(t) = e^{\gamma t} \bar{\xi}_1(t)$, $\omega'(t) = e^{\gamma t} \omega(t)$.

Form Eqs. (14), (15) and (30), the Bellman equation is yielded as follows:

$$\begin{aligned} & \rho'^{\top}(t + \delta t) P^k \rho'(t + \delta t) - \rho'^{\top}(t) P^k \rho'(t) \\ &= \int_t^{t+\delta t} \rho'^{\top}(\tau) \left((\bar{A}_{\gamma} - \bar{B} \bar{K}^k)^{\top} P^k + P^k (\bar{A}_{\gamma} - \bar{B} \bar{K}^k) \right) \rho'(\tau) d\tau \\ &+ \int_t^{t+\delta t} 2 \left(\omega'^{\top}(\tau) (\bar{K}^k)^{\top} + \xi'_1{}^{\top}(\tau) \right) \bar{B}^{\top} P^k \rho'(\tau) d\tau \\ &= - \int_t^{t+\delta t} y'^{\top}(\tau) Q_e y'(\tau) d\tau + \int_t^{t+\delta t} 2 \xi'_1{}^{\top}(\tau) R \bar{K}^{k+1} \rho'(\tau) d\tau \\ &- \int_t^{t+\delta t} \rho'^{\top}(\tau) \left((\bar{K}^k)^{\top} R \bar{K}^k + \bar{Q}_z \right) \rho'(\tau) d\tau \\ &+ \int_t^{t+\delta t} 2 \omega'^{\top}(\tau) (\bar{K}^k)^{\top} \bar{B}^{\top} P^k \rho'(\tau) d\tau \end{aligned} \quad (32)$$

where $\xi'_1(t) = e^{\gamma t} \xi_1(t)$.

Substituting Eq. (31) into (32) and dividing both sides of the equation by $e^{2\gamma t}$ yields the following expression:

$$\begin{aligned} & e^{2\gamma \delta t} \bar{\xi}_1^{\top}(t + \delta t) \bar{P}^k \bar{\xi}_1(t + \delta t) - \bar{\xi}_1^{\top}(t) \bar{P}^k \bar{\xi}_1(t) \\ &= \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \left(2 \bar{\xi}_1^{\top}(\tau) R \bar{K}_M^{k+1} \bar{\xi}_1(\tau) - y^{\top}(\tau) Q_e y(\tau) \right) d\tau \\ &- \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \bar{\xi}_1^{\top}(\tau) \left((\bar{K}_M^k)^{\top} R \bar{K}_M^k + \bar{Q}_M \right) \bar{\xi}_1(\tau) d\tau + \Omega_1^k(t) \end{aligned} \quad (33)$$

where $\bar{P}^k = \bar{M}_{\bar{\xi}}^{\top} P^k \bar{M}_{\bar{\xi}} \in \mathbb{R}^{n_{\bar{\xi}} \times n_{\bar{\xi}}}$ and $\bar{Q}_M = \bar{M}_{\bar{\xi}}^{\top} \bar{Q}_z \bar{M}_{\bar{\xi}} \in \mathbb{R}^{n_{\bar{\xi}} \times n_{\bar{\xi}}}$ with $\bar{M}_{\bar{\xi}}$ being full row rank. $\Omega_1^k(t)$ denotes the overall influence of the re-expression error $\omega(t)$ and is given by

$$\begin{aligned} \Omega_1^k(t) &= 2 \omega^{\top}(t) P^k \rho(t) - 2 e^{2\gamma \delta t} \omega^{\top}(t + \delta t) P^k \rho(t + \delta t) \\ &+ e^{2\gamma \delta t} \omega^{\top}(t + \delta t) P^k \omega(t + \delta t) - \omega^{\top}(t) P^k \omega(t) \\ &- \int_t^{t+\delta t} 2 e^{2\gamma(\tau-t)} \omega^{\top}(\tau) \left((\bar{K}^k)^{\top} R \bar{K}^k + \bar{Q}_z - (\bar{K}^k)^{\top} \bar{B}^{\top} P^k \right) \rho(\tau) d\tau \\ &+ \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \omega^{\top}(\tau) \left((\bar{K}^k)^{\top} R \bar{K}^k + \bar{Q}_z \right) \omega(\tau) d\tau \\ &+ \int_t^{t+\delta t} 2 e^{2\gamma(\tau-t)} \omega^{\top}(\tau) (\bar{K}^{k+1})^{\top} R \xi_1(\tau) d\tau \end{aligned}$$

Remark 6. Given that matrix $\bar{M}_{\bar{\xi}}$ encompasses the knowledge of system dynamics, it remains unidentified. The matrices $\bar{P}^k$ and $\bar{K}_M^k$ associated with $\bar{M}_{\bar{\xi}}$ can be resolved through iterative algorithms which will be discussed in the subsequent content. And the matrix $\bar{Q}_M$ is solely related to matrix Q_z .

Noting that $\Omega_1^k(t)$ contains an unknown vector $\omega(t)$, we have the following result.

Lemma 7. Based on Lemma 6, for a specified convergence rate criterion γ , if the vector $\rho(t)$ is bounded, $\Omega_1^k(t)$ converges to zero.

Proof. Based on the conclusions of Lemma 6, combined with the analysis of Eq. (26), it can be concluded that the re-expression errors of both the system state $x(t)$ and its associated state $x_c(t)$ exhibit asymptotic convergence to zero. Specifically, the re-expression error $\tilde{x}(t)$ of state $x(t)$ and the error $\omega(t)$ of state $x_c(t)$ satisfy $\lim_{t \rightarrow \infty} \tilde{x}(t) = 0_{n \times 1}$ and $\lim_{t \rightarrow \infty} \omega(t) = 0_{(n+pq) \times 1}$ respectively. If the vector $\rho(t)$ is bounded, then $\Omega_1^k(t)$ converges to zero. This completes the proof. $\square$

To simplify the computation of the Eq. (33), expressions $\bar{\xi}_1^{\top}(t) R \bar{K}_M^{k+1} \bar{\xi}_1(t)$ and the quadratic form such as $\bar{\xi}_1^{\top}(t) \bar{P}^k \bar{\xi}_1(t)$ are rewritten as

$$\bar{\xi}_1^{\top}(t) R \bar{K}_M^{k+1} \bar{\xi}_1(t) = (\bar{\xi}_1(t) \otimes \xi_1(t))^{\top} \left(I_{n_{\bar{\xi}}} \otimes R \right) \text{vec}(\bar{K}_M^{k+1}) \quad (34a)$$

$$\bar{\xi}_1^{\top}(t) \bar{P}^k \bar{\xi}_1(t) = \text{vec}^{\top}(\bar{\xi}_1(t)) \text{vecs}(\bar{P}^k) \quad (34b)$$

Based on Eq. (34), the data required for the k th iteration are denoted as

$$\begin{aligned}\psi_{\bar{\xi}\bar{\xi}}^k(t_j^k) &= e^{2\gamma\delta t} \text{vecv}(\bar{\xi}_1(t_{j+1}^k)) - \text{vecv}(\bar{\xi}_1(t_j^k)) \\ \Psi_{\bar{\xi}\bar{\xi}}^k &= [\psi_{\bar{\xi}\bar{\xi}}^k(t_0^k), \dots, \psi_{\bar{\xi}\bar{\xi}}^k(t_{s-1}^k)]^\top \\ \Phi_{yy}^k &= \left[\int_{t_0^k}^{t_1^k} e^{2\gamma(\tau-t_0^k)} \text{vecv}(y(\tau)) d\tau, \dots, \int_{t_{s-1}^k}^{t_s^k} e^{2\gamma(\tau-t_{s-1}^k)} \text{vecv}(y(\tau)) d\tau \right]^\top \\ \Phi_{\bar{\xi}\bar{\xi}}^k &= \left[\int_{t_0^k}^{t_1^k} e^{2\gamma(\tau-t_0^k)} \text{vecv}(\bar{\xi}_1(\tau)) d\tau, \dots, \int_{t_{s-1}^k}^{t_s^k} e^{2\gamma(\tau-t_{s-1}^k)} \text{vecv}(\bar{\xi}_1(\tau)) d\tau \right]^\top \\ \Phi_{\bar{\xi}\xi}^k &= \left[\int_{t_0^k}^{t_1^k} e^{2\gamma(\tau-t_0^k)} \bar{\xi}_1(\tau) \otimes \xi_1(\tau) d\tau, \dots, \int_{t_{s-1}^k}^{t_s^k} e^{2\gamma(\tau-t_{s-1}^k)} \bar{\xi}_1(\tau) \otimes \xi_1(\tau) d\tau \right]^\top \\ \Phi_{\Omega_1}^k &= [\Omega_1^k(t_0^k), \dots, \Omega_1^k(t_{s-1}^k)]^\top\end{aligned}$$

where t_0^k and t_s^k denote the starting and ending time, respectively, for data collection during the k -th iteration, and $t_{j+1}^k - t_j^k = \delta t$, $j = 0, \dots, s-1$.

The Bellman equation set can be obtained according to Eq. (33).

$$\theta_1^k [\text{vecs}^\top(\bar{P}^k), \text{vec}^\top(\bar{K}_M^{k+1})]^\top = \chi_1^k + \Phi_{\Omega_1}^k \quad (35)$$

where $\theta_1^k = [\Psi_{\bar{\xi}\bar{\xi}}^k, -2\Phi_{\bar{\xi}\bar{\xi}}^k(I_{n_\xi} \otimes R)]^\top$ and

$$\chi_1^k = -\Phi_{yy}^k \text{vecs}(Q_e) - \Phi_{\bar{\xi}\bar{\xi}}^k \text{vecs}((\bar{K}_M^k)^\top R \bar{K}_M^k + \bar{Q}_M).$$

Given that $\Phi_{\Omega_1}^k$ involves unknown variables, it is not feasible to compute the solution $\bar{K}_M^{k+1}$ from Eq. (35). We consider the following equation:

$$\theta_1^k [\text{vecs}^\top(\bar{P}_c^k), \text{vec}^\top(\bar{K}_{Mc}^{k+1})]^\top = \chi_{c1}^k \quad (36)$$

where $\chi_{c1}^k = -\Phi_{yy}^k \text{vecs}(Q_e) - \Phi_{\bar{\xi}\bar{\xi}}^k \text{vecs}((\bar{K}_{Mc}^k)^\top R \bar{K}_{Mc}^k + \bar{Q}_M)$, the subscript c is used to distinguish the computed result in Eq. (36) from the theoretic result in Eq. (35).

It is worth noting that, for the zero initial state re-expression error ($\omega(0) = 0_{(n+pq) \times 1}$), Eq. (36) is equal to Eq. (35). For the nonzero initial state re-expression error ($\omega(0) \neq 0_{(n+pq) \times 1}$), we obtain the following result.

Theorem 2. Under the condition that Lemma 7 is satisfied, assume that the matrix θ_1^k obtained during the time interval $[t_0^k, t_s^k]$ has a rank of $\frac{n_\xi(n_\xi+2m+1)}{2}$, and that the time t_0^k is sufficiently large (i.e., enough time has passed before data collection commenced), it can be inferred that the computed solution to Eq. (36) serves as an approximate solution to $\bar{P}^* = \bar{M}_\xi^\top P^* \bar{M}_\xi$ and $\bar{K}_M^* = \bar{K}^* \bar{M}_\xi$, with a negligible margin of error.

Proof. The proof has two parts. The first part is to show that $\bar{P}^k$ and $\bar{K}_M^{k+1}$ obtained by Eq. (35) converge to $\bar{P}^*$ and $\bar{K}_M^*$. With the rank condition $\text{rank}(\theta_1^k) = \frac{n_\xi(n_\xi+2m+1)}{2}$, Eq. (35) has an unique solution by using the least squares (LS) method. Then, $\bar{P}^k$ and $\bar{K}_M^{k+1}$ obtained by Eq. (35) must satisfy Eq. (32). This implies that P^k and K^{k+1} are equivalent to the ones in the Eqs. (14) and (15). By Lemma 2, the convergences of $\bar{P}^k$ and $\bar{K}_M^{k+1}$ are ensured.

The second part is to show that the solution to Eq. (36) serves as an approximate solution to Eq. (35). With the condition in Theorem 2 satisfied, $\Omega_1^k(t)$ converges to zero. As proven in [32, Theorem 2], if the system runs long enough before data collection begins, then the errors in the solutions of Eqs. (35) and (36) can be reduced to a negligible level.

As a result, the solution to Eq. (36) serves as an approximate solution to $\bar{P}^*$ and $\bar{K}_M^*$. The proof is completed. $\square$

The Theorem 2 implies that Eq. (35) can be approximately solved using Eq. (36), and $\bar{P}_c^k$ and $\bar{K}_{Mc}^{k+1}$ can be obtained using the method of LS.

$$[\text{vecs}^\top(\bar{P}_c^k), \text{vec}^\top(\bar{K}_{Mc}^{k+1})]^\top = ((\theta_1^k)^\top \theta_1^k)^{-1} (\theta_1^k)^\top \chi_{c1}^k \quad (37)$$

Now, the data-driven PI-based ADP algorithm for optimal output tracking control with assured convergence rate is proposed in Algorithm 1.

Remark 7. Algorithm 1 comprises two phases: the preparatory phase and the iterative phase. The preparatory phase aims to ensure that the re-expression error $\omega(t)$ is sufficiently small, which depends solely on the model of observer (17) and the initial state $x(0)$. Consequently, altering the controller for the system (1) does not impact $\omega(t)$. Moreover, control gains $\bar{K}_{Mc}^k$ ensure the stability of the system (30), which guarantees the boundedness of the vector $\rho(t)$. This boundedness satisfies the condition specified in Lemma 7. It should be noted that the minimum length of data required for each iteration in Algorithm 1 is significantly influenced by n_w , which is why we consider using a reduced-order observer.

Algorithm 1: The data-driven PI-based ADP algorithm for optimal output tracking control with assured convergence rate.

Initialization. Let $\zeta_u(0) = 0_{n_{um} \times 1}$. Select a stabilizing control gain $\bar{K}_{Mc}^0$ and a tolerance bound for the termination $\kappa > 0$. Design the pair (F, J) to satisfy the conditions specified in Lemma 4 and Lemma 5. Set the iteration index $k \leftarrow 0$.

Preparatory Phase. Apply the controller (27) with $\bar{K}_{Mc}^0$ to the system (1) until time t_0^0 .

Iterative Phase.

$$z_1(t_0^0) \leftarrow z(t_0^0), \zeta_{u_1}(t_0^0) \leftarrow \zeta_u(t_0^0).$$

loop:

Apply the controller (28) with $\bar{K}_{Mc}^k$ to the system (1), meanwhile, collect data blocks $\Psi_{\bar{\xi}\bar{\xi}}^k, \Phi_{yy}^k, \Phi_{\bar{\xi}\bar{\xi}}^k$ and $\Phi_{\bar{\xi}\bar{\xi}}^k$ for computing θ_1^k and χ_1^k .

if $\text{rank}(\theta_1^k) = \frac{n_{\bar{\xi}}(n_{\bar{\xi}}+2m+1)}{2}$ **then**
Solve Eq. (37) to obtain $\bar{K}_{Mc}^{k+1}$.

if $\|\bar{K}_{Mc}^{k+1} - \bar{K}_{Mc}^k\| < \kappa$ **then**
 $\bar{K}_{Mc}^* \leftarrow \bar{K}_{Mc}^{k+1}$
break

end

$k \leftarrow k + 1$

end

end loop

3.3. VI-Based ADP algorithm using input-output data

Algorithm 1 requires an initial stabilizing control gain $\bar{K}_{Mc}^0$ that can be difficult to obtain due to the requirement for the convergence rate. In this subsection, we utilize the VI algorithm to update matrix P , thereby bypassing the need for an initial stabilizing control gain.

From Eqs. (16) and (30), the equation can be written as follows:

$$\begin{aligned} & \rho'^\top(t + \delta t) P^k \rho'(t + \delta t) - \rho'^\top(t) P^k \rho'(t) \\ &= \int_t^{t+\delta t} \rho'^\top(\tau) \Lambda^k \rho'(\tau) d\tau - \int_t^{t+\delta t} y'^\top(\tau) Q_e y'(\tau) d\tau \\ & \quad - \int_t^{t+\delta t} \rho'^\top(\tau) \bar{Q}_z \rho'(\tau) d\tau + \int_t^{t+\delta t} 2u_1'^\top(\tau) R \bar{K}^{k+1} \rho'(\tau) d\tau \end{aligned} \quad (38)$$

where $\Lambda^k = \bar{A}_\gamma^\top P^k + P^k \bar{A}_\gamma + Q$.

Upon further expansion, we get

$$\begin{aligned} & e^{2\gamma\delta t} \bar{\xi}_1^\top(t + \delta t) \bar{P}^k \bar{\xi}_1(t + \delta t) - \bar{\xi}_1^\top(t) \bar{P}^k \bar{\xi}_1(t) \\ &= \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \bar{\xi}_1^\top(\tau) \bar{\Lambda}^k \bar{\xi}_1(\tau) d\tau - \int_t^{t+\delta t} e^{2\gamma(\tau-t)} y^\top(\tau) Q_e y(\tau) d\tau \\ & \quad - \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \bar{\xi}_1^\top(\tau) \bar{Q}_M \bar{\xi}_1(\tau) d\tau \\ & \quad + \int_t^{t+\delta t} 2e^{2\gamma(\tau-t)} u_1^\top(\tau) R \bar{K}_M^{k+1} \bar{\xi}_1(\tau) d\tau + \Omega_2^k(t) \end{aligned} \quad (39)$$

where $\bar{\Lambda}^k = \bar{M}_\xi^\top \Lambda^k \bar{M}_\xi \in \mathbb{R}^{n_{\bar{\xi}} \times n_{\bar{\xi}}}$ and $\Omega_2^k(t)$ is given by

$$\begin{aligned} \Omega_2^k(t) &= 2\omega^\top(t) P^k \rho(t) - 2e^{2\gamma\delta t} \omega^\top(t + \delta t) P^k \rho(t + \delta t) \\ & \quad + e^{2\gamma\delta t} \omega^\top(t + \delta t) P^k \omega(t + \delta t) - \omega^\top(t) P^k \omega(t) \\ & \quad + \int_t^{t+\delta t} 2e^{2\gamma(\tau-t)} \omega^\top(\tau) (\Lambda^k - \bar{Q}_z) \rho(\tau) d\tau \\ & \quad - \int_t^{t+\delta t} e^{2\gamma(\tau-t)} \omega^\top(\tau) (\Lambda^k - \bar{Q}_z) \omega(\tau) d\tau \\ & \quad + \int_t^{t+\delta t} 2e^{2\gamma(\tau-t)} \omega^\top(\tau) (\bar{K}^{k+1})^\top R u_1(\tau) d\tau \end{aligned}$$

Define

$$\Phi_{\xi u}^k = \left[\int_{t_0^k}^{t_1^k} e^{2\gamma(\tau-t_0^k)} \tilde{\zeta}_1(\tau) \otimes u_1(\tau) d\tau, \dots, \int_{t_{s-1}^k}^{t_s^k} e^{2\gamma(\tau-t_{s-1}^k)} \tilde{\zeta}_1(\tau) \otimes u_1(\tau) d\tau \right]^\top$$

$$\Phi_{\Omega_2}^k = [\Omega_2^k(t_0^k), \dots, \Omega_2^k(t_{s-1}^k)]^\top$$

Using the above notations, Eq. (39) can be rewritten as the following equation:

$$\theta_2 [\text{vecs}^\top(\bar{\Lambda}^k), \text{vec}^\top(\bar{K}_{M_c}^{k+1})]^\top = \chi_2 + \Phi_{\Omega_2}^1 \quad (40)$$

where $\theta_2 = [\Phi_{\xi e}^1, 2\Phi_{\xi u}^1(I_{n_\xi} \otimes R)]$ and $\chi_2 = \Psi_{\xi \xi}^1 \text{vecs}(\bar{P}^k) + \Phi_{yy}^1 \text{vecs}(Q_e) + \Phi_{\xi \xi}^1 \text{vecs}(\bar{Q}_M)$.

Similar to Eq. (36), we consider the following equation:

$$\theta_2 [\text{vecs}^\top(\bar{\Lambda}_c^k), \text{vec}^\top(\bar{K}_{M_c}^{k+1})]^\top = \chi_{c2} \quad (41)$$

where $\chi_{c2} = \Psi_{\xi \xi}^1 \text{vecs}(\bar{P}_c^k) + \Phi_{yy}^1 \text{vecs}(Q_e) + \Phi_{\xi \xi}^1 \text{vecs}(\bar{Q}_M)$.

Then, we have

$$[\text{vecs}^\top(\bar{\Lambda}_c^k), \text{vec}^\top(\bar{K}_{M_c}^{k+1})]^\top = (\theta_2^\top \theta_2)^{-1} \theta_2^\top \chi_{c2} \quad (42)$$

$\bar{\Lambda}_c^k$ and $\bar{K}_{M_c}^k$ can be computed by solving Eq. (42), and subsequently used to obtain $\bar{P}_c^{k+1}$ through the equation derived from Eq. (16).

$$\begin{cases} \bar{P}_c^{k+1} = \bar{P}_c^k + \varepsilon^k (\bar{\Lambda}_c^k - (\bar{K}_{M_c}^k)^\top R \bar{K}_{M_c}^k) \\ \bar{P}_c^{k+1} = \begin{cases} \bar{P}_c^0 & \bar{P}_c^{k+1} \notin B^i \\ \bar{P}_c^{k+1} & \bar{P}_c^{k+1} \in B^i \end{cases} \end{cases} \quad (43)$$

where ε^k and $\{B^i\}_{i=0}^\infty$ are mentioned in Lemma 3.

The data-driven VI-based ADP algorithm for optimal output tracking control with assured convergence rate is given in Algorithm 2.

The convergence analysis of Algorithm 2 is presented in Theorem 3. The proofs of Lemma 8 and Theorem 3 are analogous to those of Lemma 7 and Theorem 2, respectively, and are therefore omitted.

Lemma 8. Based on Lemma 6, for a specified convergence rate criterion γ , if the vector $\rho(t)$ is bounded, $\Omega_2^k(t)$ converges to zero.

Theorem 3. Under the condition that Lemma 8 is satisfied, assume that the matrix θ_2 obtained during the time interval $[t_0^1, t_s^1]$ has a rank of $\frac{n_\xi(n_\xi+2m+1)}{2}$, and that the time t_0 is sufficiently large, it can be inferred that the computed solution to Algorithm 2 serves as an approximate solution to $\bar{P}^*$ and $\bar{K}_M^*$, with a negligible margin of error.

Remark 8. Algorithm 2 eliminates the necessity for an initial stabilizing control gain for the system (30), but it also cause the state $\rho(t)$ of the system (29) to diverge, thereby violating the conditions stipulated in Lemma 8. One approach is to design a bound of $\rho(t)$, such that when collecting data, if the state $\rho(t)$ exceeds this bound, $\rho(t)$ is reset to $\rho(0)$ and data collection continues [19]. However, this approach is not applicable to Algorithm 2. The presence of the preparatory phase means that resetting the $\rho(t)$ impacts the re-expression error $\omega(t)$, leading to the failure of Algorithm 2. In response to this, we suggest designing the initial control gain in the form of $\bar{K}_{M_c}^0 = [K_y, 0_{m \times n_\omega(m+1)}, K_z]$ with $K_y \in \mathbb{R}^{m \times p}$. This may still prove unstable for the system (30), but it can facilitates stabilizing the system (29) easily to avoid divergence [40, Proposition 4.2].

4. Simulation

We validate the proposed algorithms on the short-period approximate dynamics model of the F-16 aircraft system. The system state is $x = [x_1, x_2, x_3]^\top = [\alpha, q, \delta_e]^\top$, where α , q , and δ_e represent the angle of attack, pitch rate, and elevator deflection angle, respectively. The system output is the pitch rate. Fig. 1 shows the rate and angles used in pitch motion of the aircraft. The nominal parameters corresponding to the system (1) are

$$A = \begin{bmatrix} -1.01887 & 0.90506 & -0.00215 \\ 0.82225 & -1.07741 & -0.17555 \\ 0 & 0 & -20.2 \end{bmatrix}$$

$$B = [0 \quad 0 \quad 20.2]^\top \quad C = [0 \quad 57.2958 \quad 0]$$

The expected reference trajectory of the pitch rate is generated by the external system (2), with model parameters

$$D = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad E = [-1 \quad 0]$$

Algorithm 2: The data-driven VI-based ADP algorithm for optimal output tracking control with assured convergence rate.

Initialization. Let $\zeta_u(0) = 0_{n_w m \times 1}$. Select a control gain $\bar{K}_{Mc}^0$, a tolerance bound for the termination $\kappa > 0$ and $\bar{P}_c^0 = (\bar{P}_c^0)^\top > 0$. Select proper sets $\{B^i\}_{i=0}^\infty$ and $\{\varepsilon^k\}_{k=0}^\infty$ that satisfy Lemma 3. Design the pair (F, J) to satisfy the conditions specified in Lemma 4 and Lemma 5. Set $k \leftarrow 0, i \leftarrow 0$.

Preparatory Phase. Apply the controller (27) with $\bar{K}_{Mc}^0$ to the system (1) until time t_0^0 .

Iterative Phase.

$$z_1(t_0^0) \leftarrow z(t_0^0), \zeta_{u_1}(t_0^0) \leftarrow \zeta_u(t_0^0).$$

Apply the controller (28) with $\bar{K}_{Mc}^0$ to the system (1) and collect data blocks $\Psi_{\zeta\bar{\zeta}}^1, \Phi_{yy}^1, \Phi_{\zeta\bar{\zeta}}^1$ and $\Phi_{\bar{\zeta}u}^1$ for computing θ_2 and χ_2 .

if $\text{rank}(\theta_2) = \frac{n_\zeta(n_\zeta + 2m + 1)}{2}$ then

loop:

Solve Eq. (42) and Eq. (43) to obtain $\bar{\Lambda}_c^k, \bar{K}_{Mc}^{k+1}$ and $\bar{P}_c^{k+1}$.

if $\bar{P}_c^{k+1} \notin B^i$ then

$$\bar{P}_c^{k+1} \leftarrow \bar{P}_c^0, i \leftarrow i + 1.$$

else if $\frac{\|\bar{P}_c^{k+1} - \bar{P}_c^k\|}{\varepsilon^k} < \kappa$ then

$$\bar{K}_{Mc}^* \leftarrow \bar{K}_{Mc}^{k+1}$$

break

else

$$\bar{P}_c^{k+1} \leftarrow \bar{P}_c^{k+1}$$

end

$$k \leftarrow k + 1$$

end loop

end

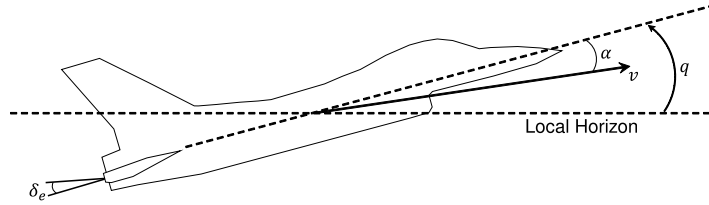


Fig. 1. The pitch movement of the aircraft.

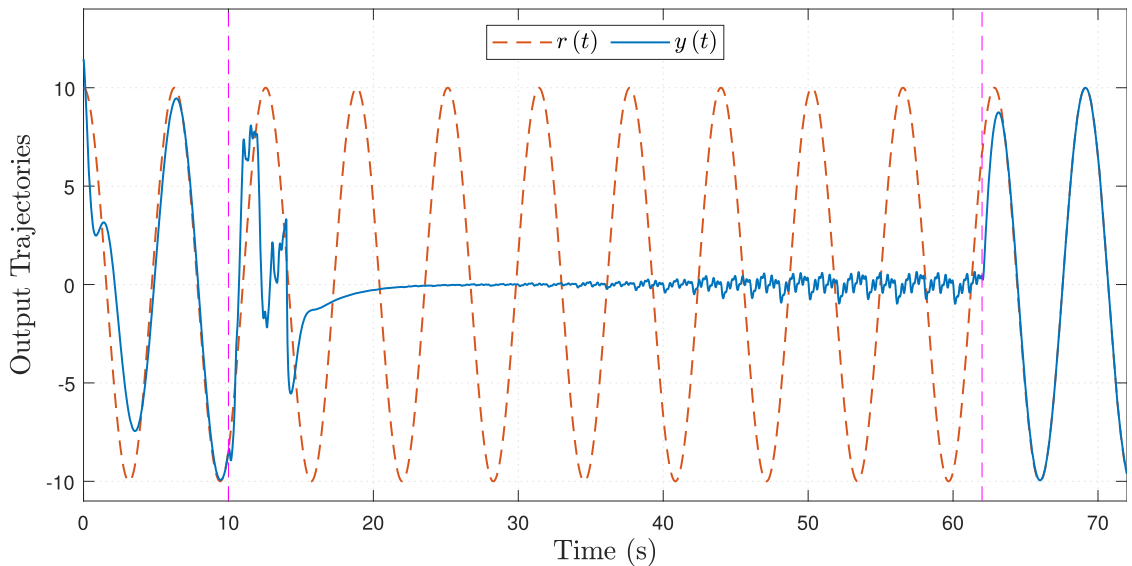


Fig. 2. The output $y(t)$ and the reference trajectory $r(t)$ via Algorithm 1.

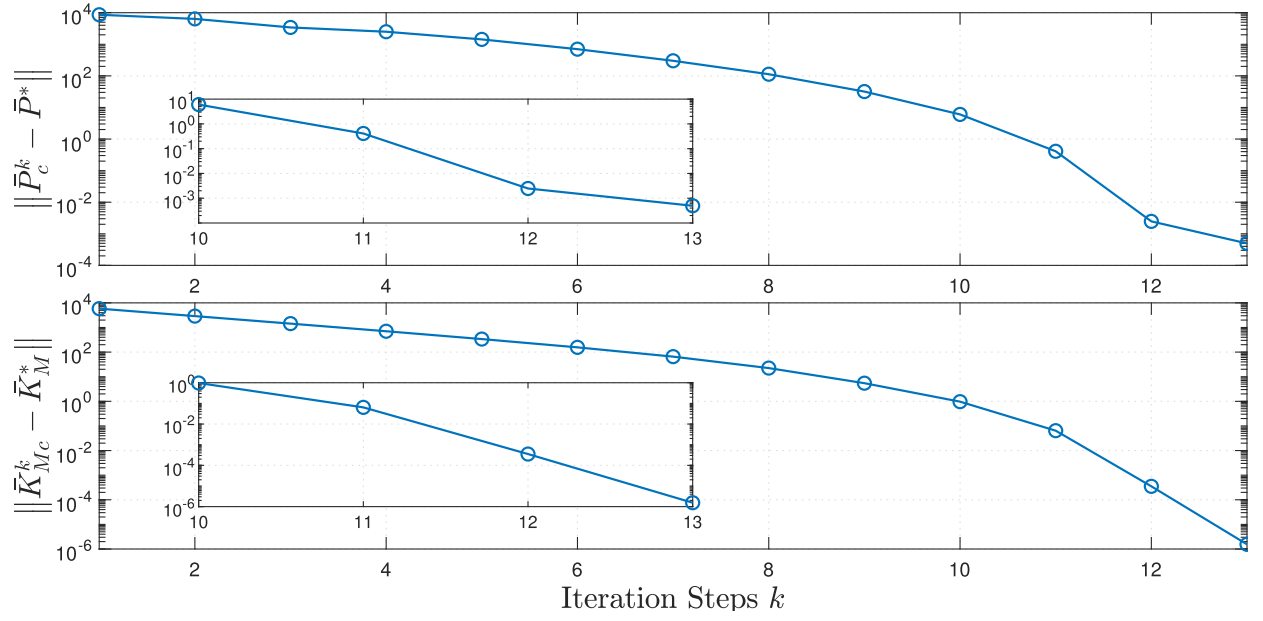


Fig. 3. The trajectories of $\|\bar{P}_c^k - \bar{P}^*\|$ and $\|\bar{K}_{M_c}^k - \bar{K}_M^*\|$ via Algorithm 1.

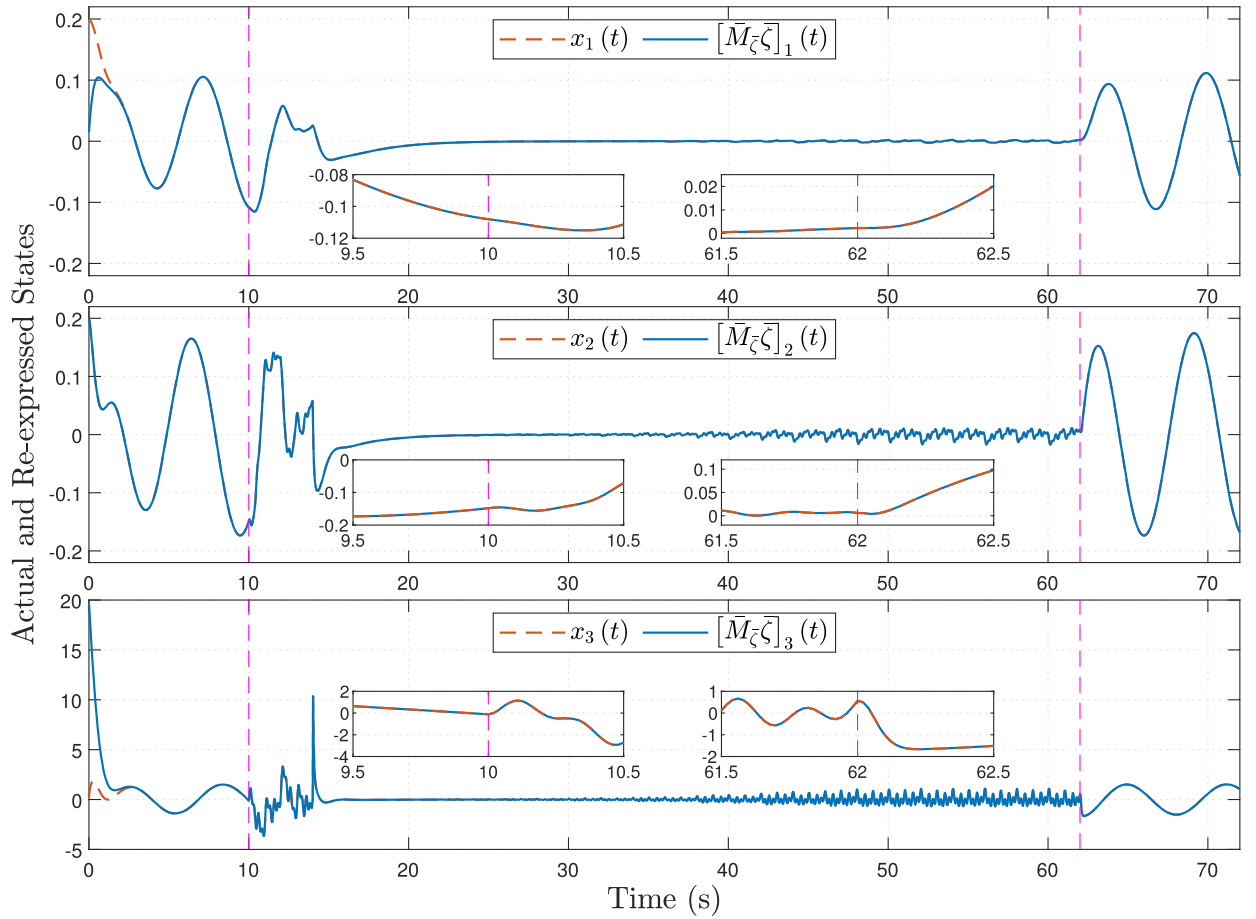


Fig. 4. The trajectories of actual and re-expressed states via Algorithm 1.

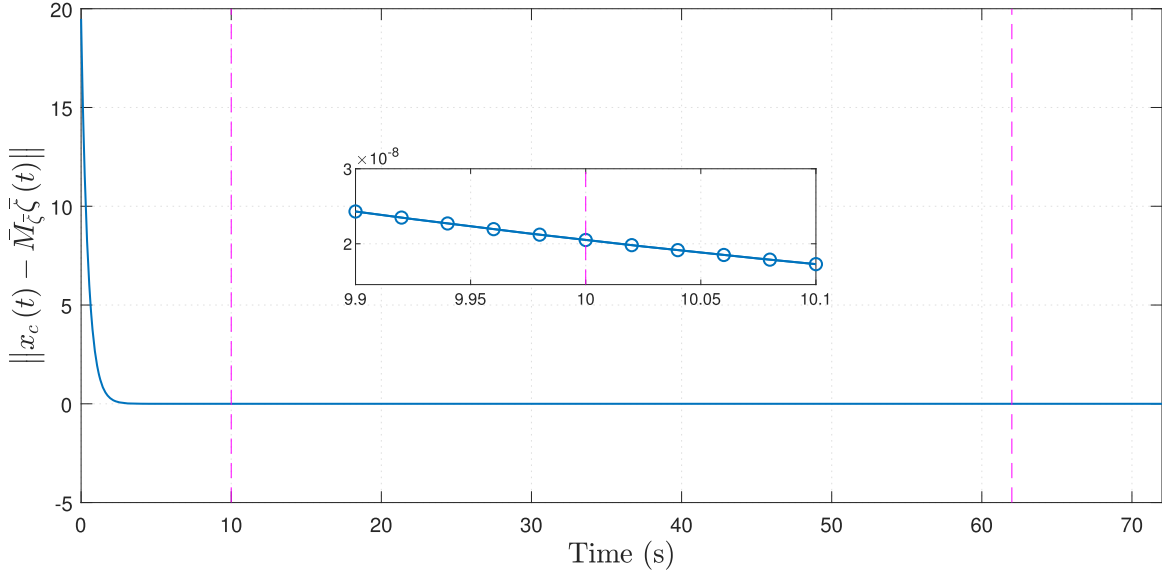


Fig. 5. The trajectories of $\|x_c(t) - \bar{M}_z \bar{z}(t)\|$ via Algorithm 1.

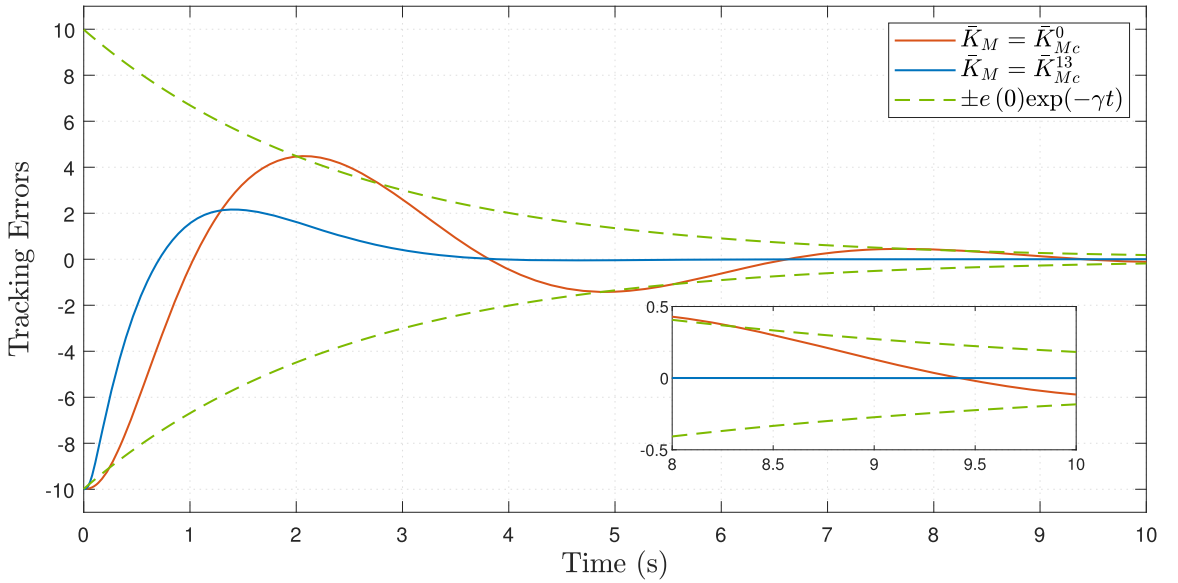


Fig. 6. Compared tracking errors with $\bar{K}_{Mc}^0$ and $\bar{K}_{Mc}^{13}$.

The minimal p -copy internal model of matrix D is designed as

$$G_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad G_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

We select the performance index parameters as $Q_e = 10$, $R = 1$ and $Q_z = 10I_2$. The matrix P^* can be obtained by solving the ARE (13) with $\gamma = 0.4$. The optimal control gain $\bar{K}^*$ is obtained from Eq. (12).

The eigenvalues of the observer (17) matrix F are set to -2 , the matrix J is set to $J = [0, 1]^T$ to satisfy the controllability of the pair (F, J) . Then, the matrix A_ζ is obtained according to the definition.

$$A_\zeta = F = \begin{bmatrix} 0 & 1 \\ -4 & -4 \end{bmatrix}$$

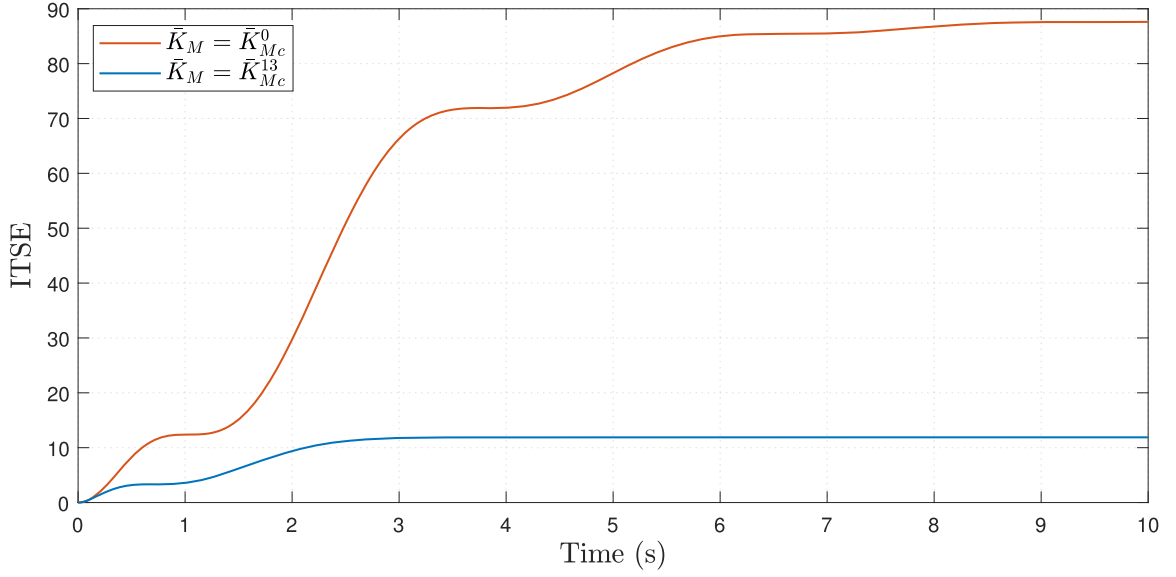


Fig. 7. Compared integral time square errors (ITSE) with $\bar{K}_{Mc}^0$ and $\bar{K}_{Mc}^{13}$.

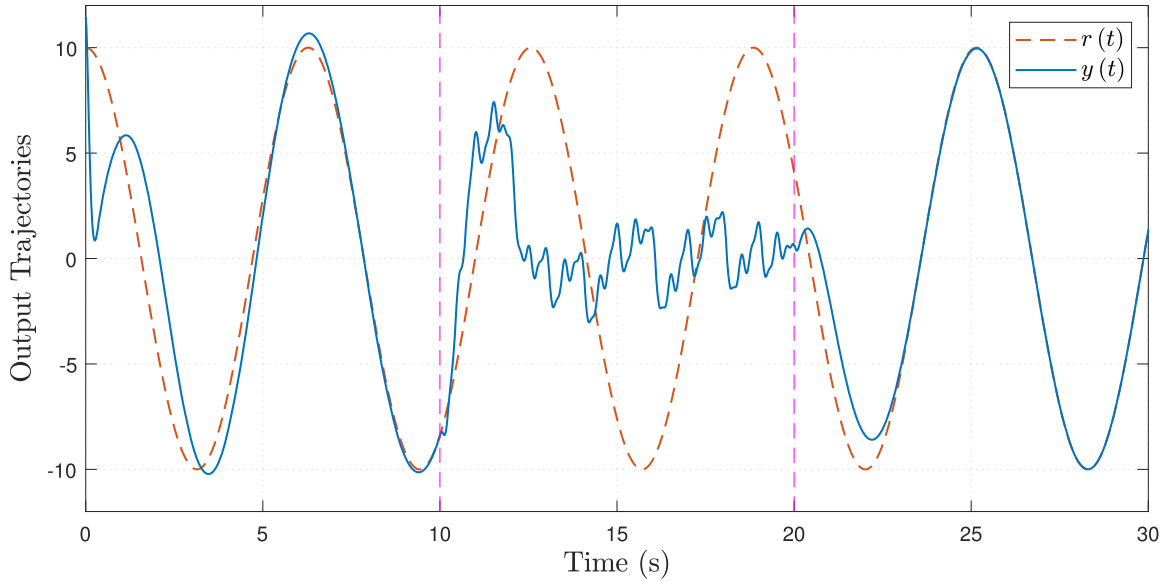


Fig. 8. The output $y(t)$ and the reference trajectory $r(t)$ via Algorithm 2.

Then, the matrix $\bar{M}_{\xi}$ is obtained from Eq. (26). After solving for $\bar{K}^*$ and $\bar{M}_{\xi}$, the optimal control gain $\bar{K}_M^*$ is obtained as

$$\bar{K}_M^* = \begin{bmatrix} -1.58217 & -7.72120 & -4.03252 & 22.60441 \\ 22.87808 & -3.95990 & -7.97925 & \end{bmatrix}$$

4.1. The data-driven PI-based ADP algorithm

For Algorithm 1, we set the initial control gain as $\bar{K}_{Mc}^0 = [-0.1, -1.6, -0.8, 1.6, 1.6, -0.2, -0.4]$. The initial values for the system state $x(t)$, the exosystem state $v(t)$, the internal model state $z(t)$, and the observer state $w_1(t)$ are given as $x(0) = [0.2, 0.2, 0.2]^T$, $v(0) = [10, 0]^T$, $z(0) = [0, 0]^T$, and $w_1(0) = [0, 0]^T$, respectively.

The sampling interval is set to 0.02s, and the duration of the preparatory phase is $t_0^0 = 10$ s. During the iterative phase, we set the minimum data collection time for each iteration to 4s. The tolerance bound for the termination is set as $\kappa = 10^{-3}$. The exploration noise is designed as $\xi_1(t) = \sum_{-2 \leq i \leq 2} \sin(2^{i+1}\pi t)$, $i \in \mathbb{Z}$.

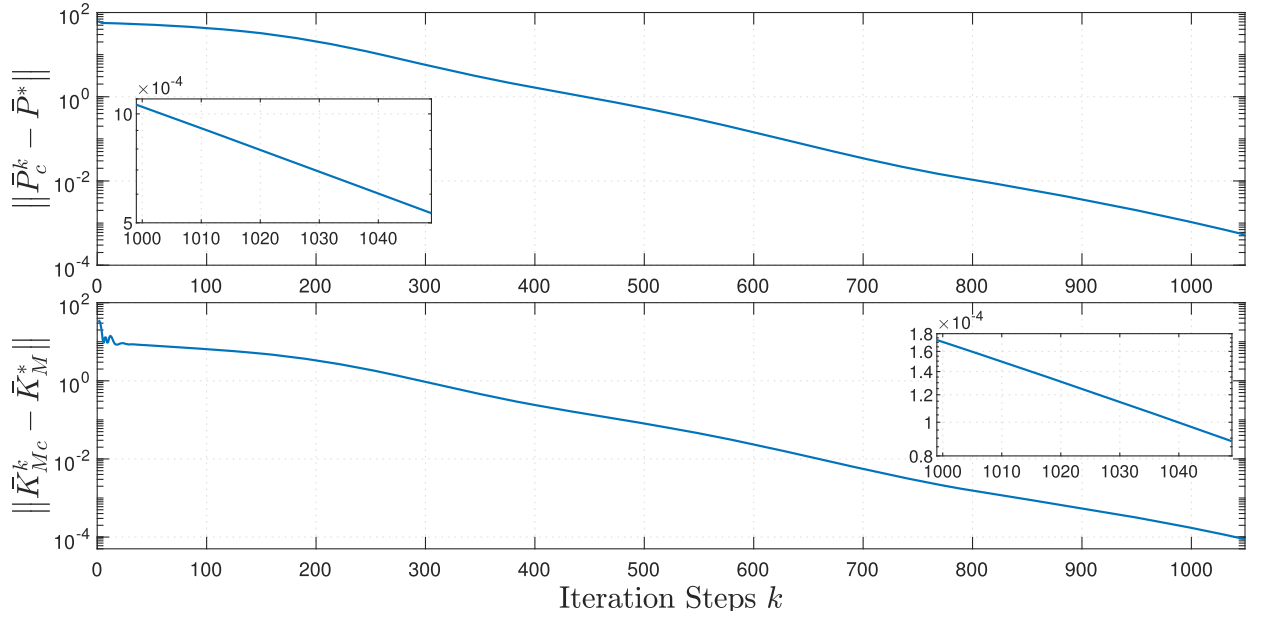


Fig. 9. The trajectories of $\|\bar{P}_c^k - \bar{P}^*\|$ and $\|\bar{K}_{Mc}^k - \bar{K}_M^*\|$ via Algorithm 2.

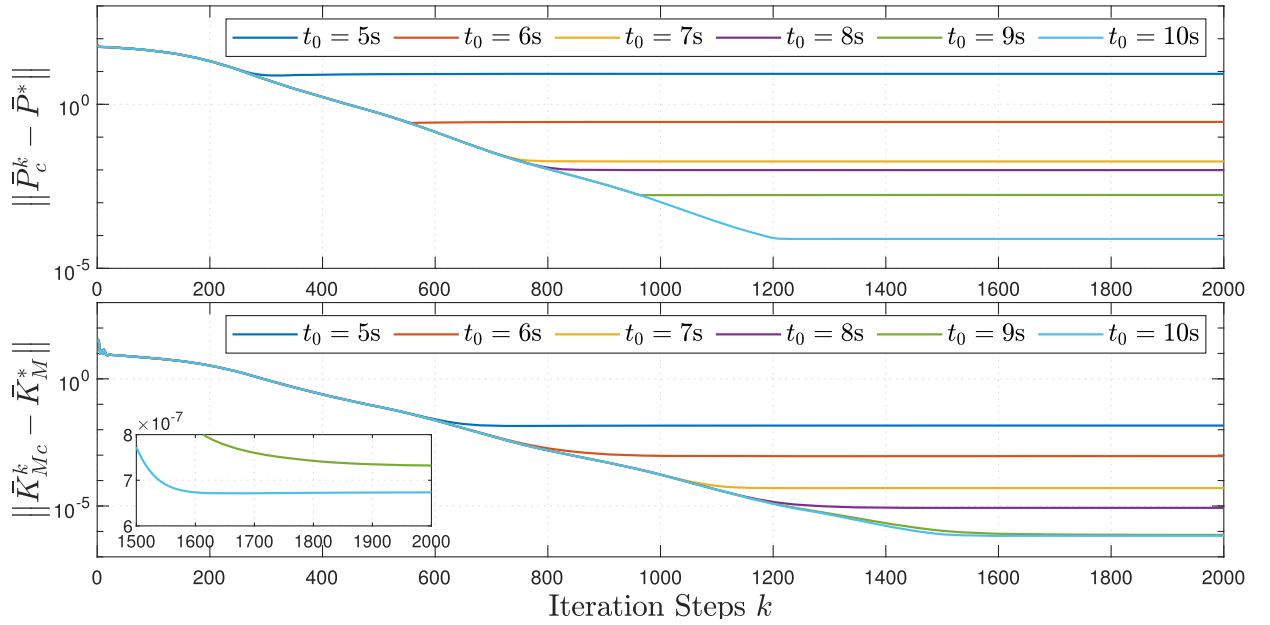


Fig. 10. Trajectories of $\|\bar{P}_c^k - \bar{P}^*\|$ and $\|\bar{K}_{Mc}^k - \bar{K}_M^*\|$ under different t_0 via Algorithm 2.

$$\bar{K}_{Mc}^{13} = \begin{bmatrix} -1.58217 & -7.72120 & -4.03251 & 22.60441 \\ 22.87808 & -3.95990 & -7.97925 & \end{bmatrix}$$

Fig. 2 shows the reference trajectory $r(t)$ and the output trajectory $y(t)$ under Algorithm 1, where the period from 0s to 10s is the preparatory phase, 10s to 62s is the iterative phase, and 62s to 72s is the output trajectory running with the control gain obtained through 13 iterations.

Comparing $\bar{K}_M^*$ and $\bar{K}_{Mc}^{13}$, it can be seen that the result obtained through iteration is very close to $\bar{K}_M^*$, which demonstrates the effectiveness of Algorithm 1. Fig. 3 provides the trajectories of $\|\bar{K}_{Mc}^k - \bar{K}_M^*\|$ and $\|\bar{P}_c^k - \bar{P}^*\|$, from which the convergence can be better observed.

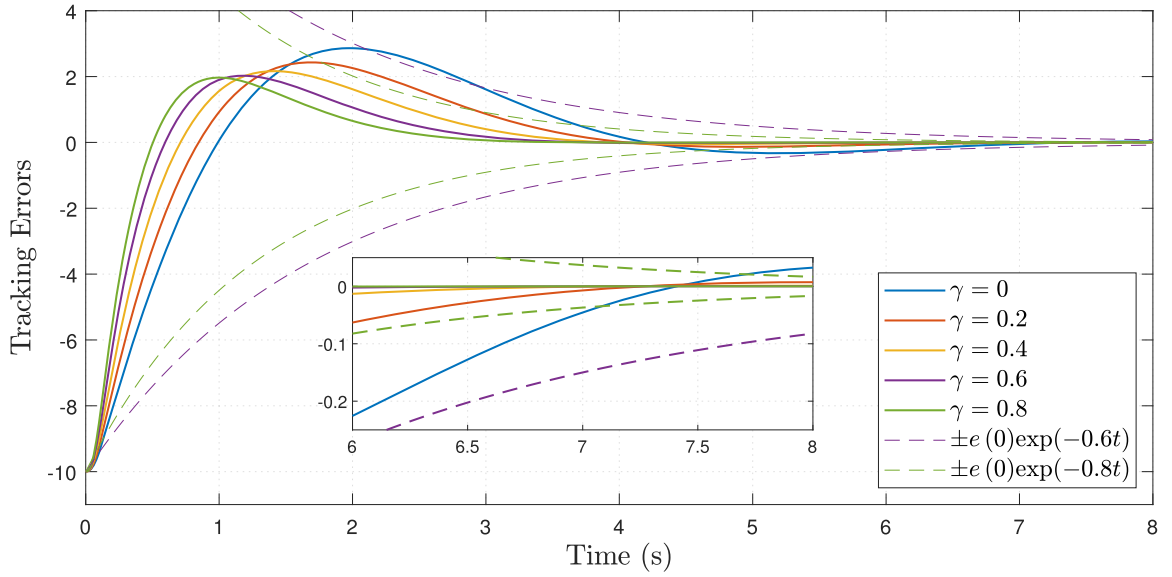
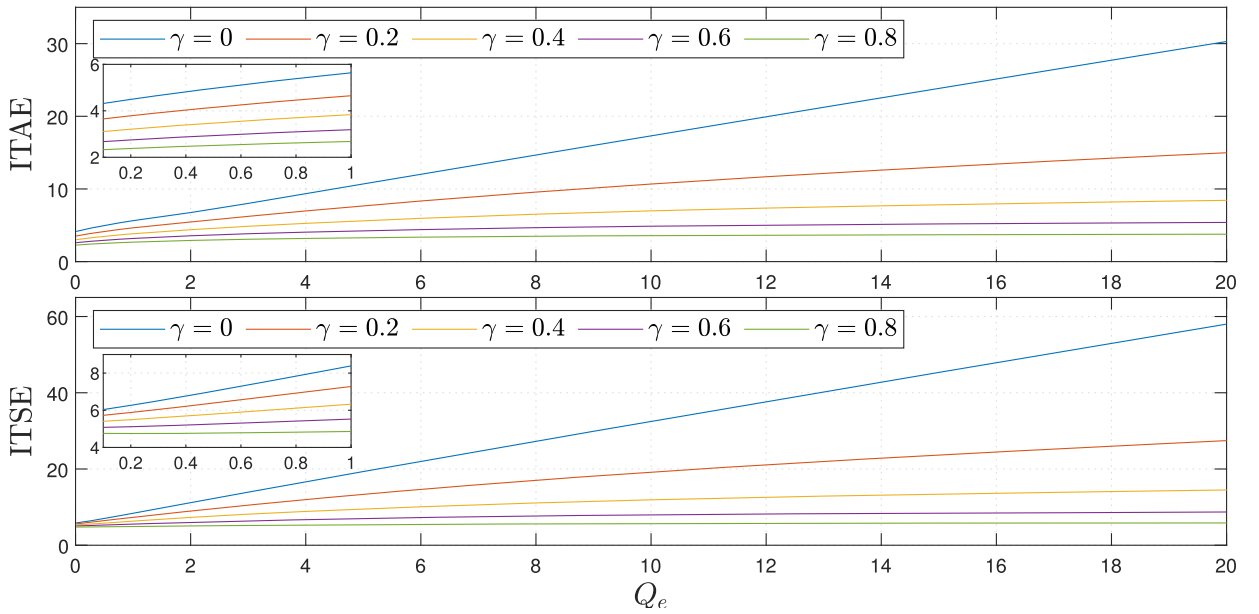
Fig. 11. Compared tracking errors with varying γ .Fig. 12. Compared ITAE and ITSE with varying γ and Q_e .

Fig. 4 presents the trajectories of the actual state and the re-expressed state. Fig. 5 presents the trajectory of $\|x_c(t) - \bar{M}_{\bar{\xi}}\bar{\xi}(t)\|$. It can be observed that the error between the actual state and the re-expressed error state converges to zero within a short time, and convergence is maintained during the iterative phase. Additionally, changing the controller at the beginning and end of the iterative phase did not affect this error.

Figs. 6 and 7 illustrate the tracking errors and the integral time square errors of the system under the initial control gain and the control gain after iteration. To reduce the influence of state re-expression error on these comparison results, the system was run for a 10-second preparatory phase prior to the comparison. As shown in Figs. 6 and 7, the control gain after iteration leads to improved tracking performance for the system.

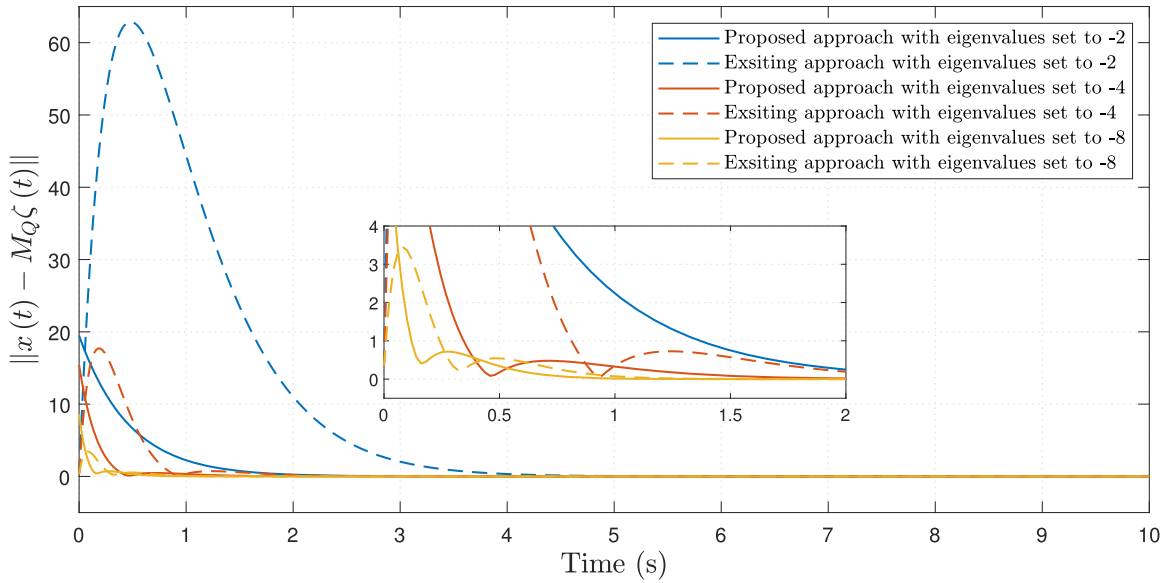


Fig. 13. Compared the norm of re-expression error with the approach in [29].

4.2. The data-driven VI-based ADP algorithm

For Algorithm 2, the parameters are chosen as follows: $\varepsilon^k = \frac{1}{100+k^{0.5}}$, $B^i = \{P > 0 | \|P\| < 100(i+1)\}$, $t_0 = 10s$, $\bar{K}_{Mc}^0 = [-1, 0, 0, 0, -1, -1]$ and $\bar{P}_c^0 = 0.5I_7$. The minimum data collection time is set to 10s. The other parameters are carried over from this section. The reference trajectory $r(t)$ and the output trajectory $y(t)$ under Algorithm 2 are shown in Fig. 8.

The result obtained after 1049 iterations using Algorithm 2 is

$$\bar{K}_{Mc}^{1049} = \begin{bmatrix} -1.58216 & -7.72119 & -4.03251 & 22.60439 \\ 22.87806 & -3.95984 & -7.97919 & \end{bmatrix}$$

By comparing $\bar{K}_M^*$ and $\bar{K}_{Mc}^{1049}$ and observing the trajectories shown in Fig. 9, the convergence of Algorithm 2 can be guaranteed.

4.3. Comparative experiments

Since Algorithm 2 employs the same set of data collected during (t_0, t_s) at each iteration, and the duration of the preparation phase t_0 significantly influences the re-expression error $\omega(t)$, which further affects the convergence performance of Algorithm 2. To evaluate this, we compared computational results under varying t_0 . As shown in Fig. 10, smaller t_0 yields larger $\|\bar{K}_{Mc}^k - \bar{K}_M^*\|$, indicating more pronounced effects of $\omega(t)$. Conversely, the impact diminishes with larger t_0 and eventually becomes negligible.

The compared simulation results with varying convergence rate criteria are depicted in Fig. 11. As the convergence rate criterion γ increases, the tracking errors converge more rapidly. Fig. 12 shows the ITAE and ITSE under varying γ and Q_e . The introduction of γ effectively suppresses the sensitivity of tracking performance to Q_e . Under a constant Q_e , increasing γ can improve the tracking performance.

Finally, we compare our state re-expression method with that presented in [29]. The comparative simulation results are presented in Fig. 13, it can be seen that the approach proposed in this paper achieves a faster convergence rate under the same eigenvalues. Furthermore, the proposed approach reduces the dimensionality of the re-expressed state by $(p-1)n + pm$.

5. Conclusions

This paper addressed the optimal output tracking problem for CT linear time-invariant systems, with a particular focus on scenarios where the system dynamics and state are unknown and a specified convergence rate for the tracking error is required. The primary contribution of this study was to propose a method based on the reduced-order observer for re-expressing the state vector using input-output data. The proposed state re-expression method effectively reduced the dimension of the re-expressed state compared to existing methods. Moreover, a dynamic output feedback controller was designed, and two ADP-based algorithms were introduced in conjunction with PI and VI algorithms, respectively. These approaches eschewed reliance on the system dynamics and state information, instead utilizing online data to obtain the optimal control gain and ensure convergence rate with less data required compared to existing algorithms. Our future work is to design data-driven algorithms for adaptive optimal output tracking control of unknown nonlinear systems with a specified convergence rate of tracking error.

CRediT authorship contribution statement

Shenhao Yang: Writing – original draft, Software, Methodology, Conceptualization; **Huarong Zhao:** Writing – review & editing, Methodology, Investigation, Funding acquisition; **Masaki Ogura:** Writing – review & editing, Visualization; **Li Peng:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author, Masaki Ogura, is an Associate Editor for the Journal of The Franklin Institute and was not involved in the editorial review or the decision to publish this article.

Acknowledgment

This work was supported in part by The [National Natural Science Foundation of China](#) under Grant 62403216, in part by the [Basic Research Program of Jiangsu Province](#) under Grant BK20241608, in part by the Japan Science and Technology Agency Moonshot R&D under Grant JPMJMS2389, in part by the Wuxi Science and Technology Development Fund Project under Grant K20231015, in part by the Wuxi Young Science and Technology Talent Support Program under Grant TJXD-2024-114, and in part by the 111 Project under Grant B23008.

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